Post-Earnings Announcement Drift in S&P 500 Companies: A Bootstrap Analysis of Variance Changepoints

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2026-04-21

Introduction

This project evaluates whether financial markets fully incorporate earnings information at the time of announcement or whether a measurable post-announcement adjustment window persists. The analysis focuses on stock return behavior for S&P 500 companies from 2015 to 2025, using daily return data aligned with SEC filing dates. Each filing event generates a 31-day return window spanning from the filing date (Day 0) through Day 30.

The central question is whether price adjustment occurs instantaneously, as predicted by the Efficient Market Hypothesis, or whether the market requires additional time to process and incorporate new information. More specifically, this project seeks to identify where, within the post-announcement window, the structure of returns transitions from an initial high-uncertainty regime to a more stable state.

To address these questions, the analysis adopts a functional data framework, treating each filing event as a return trajectory over time rather than as a collection of independent observations. The primary focus is on the behavior of the variance function across the event window. To characterize this behavior, the analysis combines changepoint detection using the Inclan-Tiao method with a bootstrap-based testing procedure that compares early and later segments of the return window.

The results indicate that while markets adjust rapidly to new information, adjustment is not instantaneous. Instead, the majority of variance resolution occurs within the first three to four trading days following an earnings announcement, suggesting the presence of a short but measurable post-announcement adjustment window.

Methods

Functional Data Framework

Each filing event is represented as a function $X_i(t)$, where $t \in \{0, 1, \ldots, 30\}$ denotes the number of days relative to the filing date. This formulation preserves the temporal structure of post-announcement returns and allows the dataset to be analyzed as a collection of curves.

Within this framework, the primary quantities of interest are the population-level mean and variance functions:

$$\mu(t) = E[X(t)], \quad \sigma^2(t) = \mathrm{Var}[X(t)]$$

This approach is particularly well-suited to event studies, where the objective is to understand how return behavior evolves across time rather than at any single point.

Variance Changepoint Detection

To evaluate how return variability is distributed across the event window, the analysis applies the Inclan-Tiao cumulative sum of squares method. Let $\hat{\sigma}^2(k)$ denote the estimated variance at day k. The cumulative variance is defined as:

$$C_k = \sum_{t=1}^{k} \hat{\sigma}^2(t)$$

Under the null hypothesis of constant variance, the proportion of total variance accumulated by time k should be approximately k/T, where $T = 31$. Deviations from this expectation are captured by the centered statistic:

$$D_k = \frac{C_k}{C_T} - \frac{k}{T}$$

Systematic positive values of D_k indicate that variance is concentrated in the early portion of the event window. The estimated changepoint is defined as:

$$\hat{k} = \arg\max_k |D_k|$$

This provides a data-driven estimate of where the variance structure transitions from elevated to stabilized behavior.

Bootstrap Testing Procedure

To complement the changepoint analysis, a bootstrap procedure is used to test for differences between early and later segments of the event window. For each candidate split point k, the data are partitioned into an early window (Days 0 through k) and a later window (Days $k+1$ through 30).

A test statistic measuring the difference in variability between these segments is computed and compared against a bootstrap distribution generated through repeated resampling. This produces an empirical p-value and a corresponding decision regarding whether the two segments differ significantly.

By repeating this procedure across multiple split points, the analysis identifies where the distinction between early and later return behavior is most pronounced.

To assess whether variability in stock returns differs between early and later post-announcement periods, we implement a nonparametric bootstrap procedure comparing variances across time windows.

Data Structure

Each earnings announcement event is treated as a functional observation. Let

$$X_i \ = (X_{i0} \ , X_{i1} \ , ..., X_{i30} \ ), \quad i = 1, ..., n$$

denote the sequence of percent returns for event i, where X_{it} represents the return on day t relative to the filing date.

Window Partitioning

To investigate how variability evolves over time, we partition the time domain into two groups based on a split point k:

- Early-period returns: $G_1^{(k)} = X_{i0} \ , X_{i1} \ , ..., X_{ik}$

- Late-period returns: $G_2^{(k)} = X_{i,k+1}, ..., X_{i30}$

We repeat this analysis across multiple values of k, effectively testing splits such as:

- Days 0 vs. 1–30
- Days 0–1 vs. 2–30
- Days 0–2 vs. 3–30
- etc.

Test Statistic

For each split k, we define the observed difference in variance between the two groups as:

$$T_{obs}^{(k)} = | Var(G_1^{(k)}) - Var(G_2^{(k)})|$$

This statistic measures the magnitude of the difference in variability between early and later periods.

Bootstrap Procedure

To approximate the null distribution of the test statistic, we perform the following bootstrap procedure:

1. Combine all observations from both groups into a single pooled sample: $G^{(k)} = G_1^{(k)} \cup G_2^{(k)}$
2. For $b = 1, ..., B$ bootstrap iterations:
    - Sample with replacement from $G^{(k)}$
    - Randomly partition the resampled data into two groups of sizes $| G_1^{(k)} |$ and $| G_2^{(k)} |$
    - Compute the bootstrap test statistic: $T_b^{(k)} = Var(G_{1,b}^{(k)}) - Var(G_{2,b}^{(k)})$

This procedure generates an empirical null distribution of variance differences under the assumption that both groups arise from the same underlying distribution.

P-value Calculation

The bootstrap p-value is calculated as:

$$p^{(k)} = \frac{1}{B} \sum_{b=1}^{B} \mathbf{1}\left(T_b^{(k)} \geq T_{\text{obs}}^{(k)}\right)$$

where $\mathbf{1}(\cdot)$ is the indicator function.

Decision Rule

For each split k, we compare the p-value to a significance level $\alpha = 0.05$:

- If $p^{(k)} < \alpha$, we reject the null hypothesis and conclude that the variance differs between early and late periods.
- Otherwise, we fail to reject the null hypothesis.

Interpretation

This procedure allows us to systematically evaluate whether early post-announcement returns exhibit higher variability than later periods. By examining multiple split points, we can assess whether volatility decreases abruptly (consistent with a changepoint) or more gradually over time.

Data Analysis

Mean Function

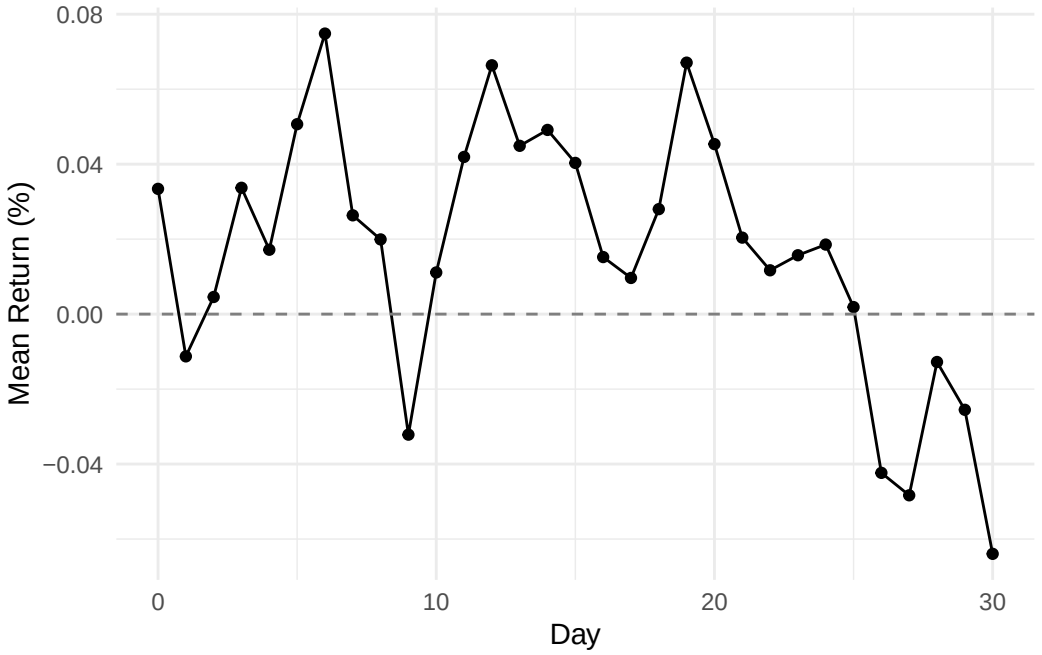


Figure 1: Estimated mean daily return (%) as a function of day relative to SEC filing date. S&P 500 companies, 2015-2025. Day 0 = filing date.

The estimated mean return function (Figure 1) remains close to zero across the entire 31-day

event window, with only minor fluctuations around the baseline. There is no sustained upward or downward trend in the post-announcement period.

This result is consistent with expectations. Because the dataset pools events associated with both positive and negative earnings disclosures, directional effects largely cancel out in the aggregate. As a result, the mean function does not provide evidence of systematic drift in this setting.

Variance Function

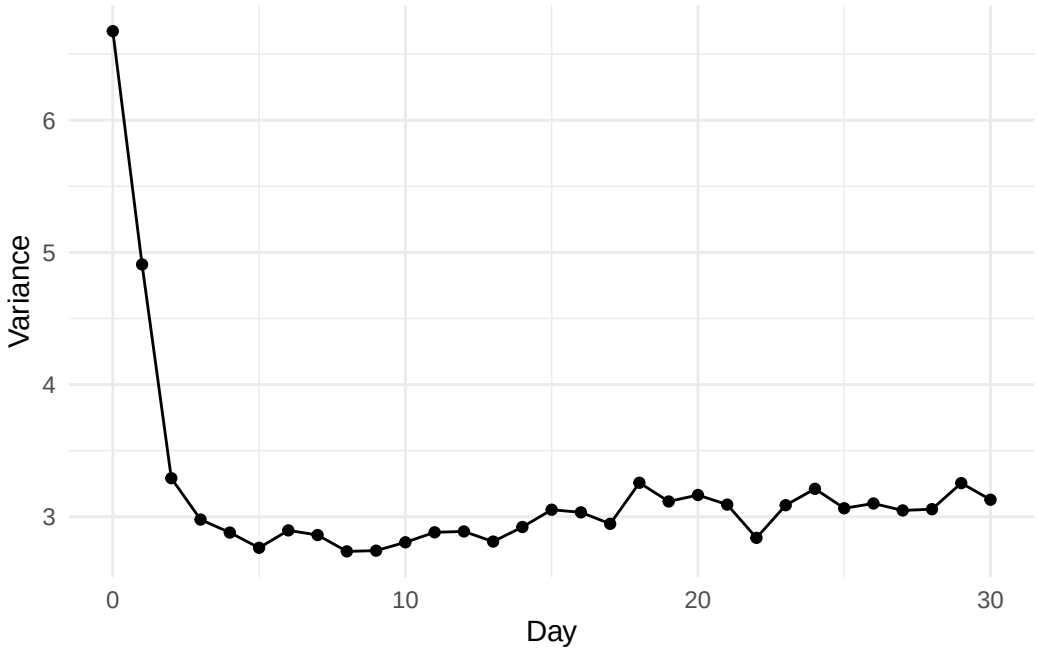


Figure 2: Estimated variance of daily returns as a function of day relative to SEC filing date. S&P 500 companies, 2015-2025. Day 0 = filing date.

In contrast to the mean, the variance function (Figure 2) exhibits a clear and structured pattern. Return variance is sharply elevated on the filing date and declines rapidly over the first several trading days before stabilizing at a lower level.

This behavior indicates that the uncertainty introduced by an earnings announcement is not resolved instantaneously. Instead, the market undergoes a short adjustment period during which volatility is elevated, followed by a transition to a more stable state. The concentration of variance in the early portion of the window provides direct evidence of this adjustment process.

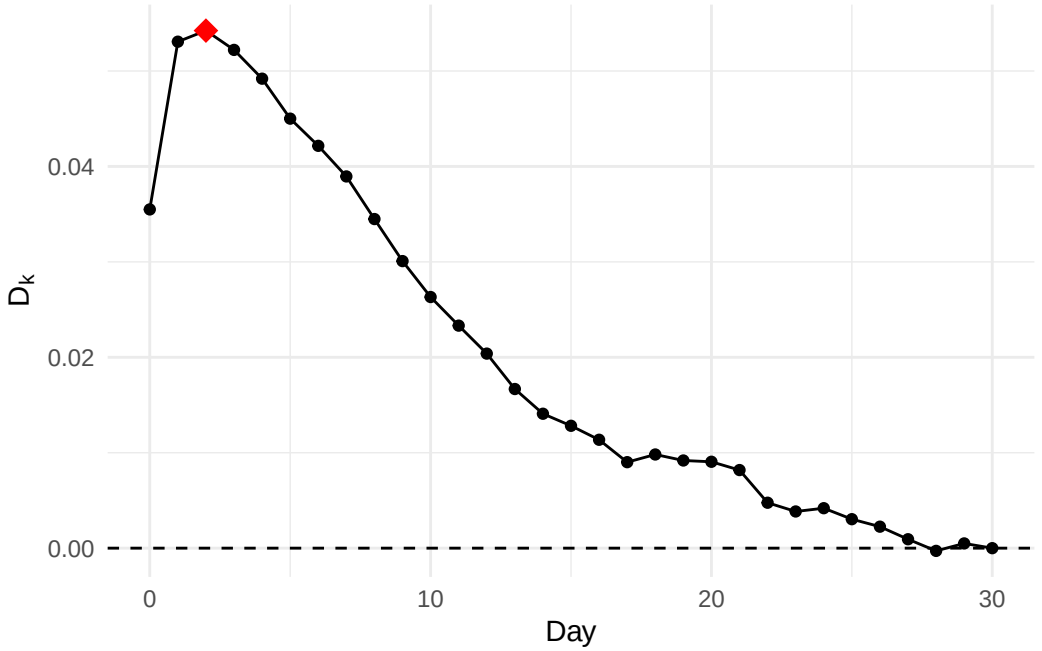


Figure 3: Centered cumulative sum of squares (D_k) as a function of day relative to SEC filing date. S&P 500 companies, 2015-2025. Diamond marker indicates estimated changepoint. Dashed line at zero is the null-hypothesis benchmark.

Changepoint Detection

The changepoint analysis reinforces this interpretation. The centered cumulative variance statistic D_k (Figure 3) reaches its maximum in the early portion of the event window, indicating that variance is disproportionately concentrated in the first few days following the filing.

This result identifies a clear transition from an initial high-variability regime to a stabilized pattern of returns. The location of the changepoint suggests that the most substantial adjustment occurs within the first two to three trading days.

Bootstrap Results

Table 1: Bootstrap permutation test results for all possible split points. P-values indicate the probability of observing a variance ratio as extreme as the observed value under the null hypothesis of equal variance structure across windows.

Slice	Group 1	Group 2	Variance Ratio	P-value	Decision
0	Day 0	Day 1-30	2.1803	0.010	Reject
1	Day 0-1	Day 2-30	1.9322	0.005	Reject

Table 1: Bootstrap permutation test results for all possible split points. P-values indicate the probability of observing a variance ratio as extreme as the observed value under the null hypothesis of equal variance structure across windows.

Slice	Group 1	Group 2	Variance Ratio	P-value	Decision
2	Day 0-2	Day 3-30	1.6601	0.005	Reject
3	Day 0-3	Day 4-30	1.4942	0.000	Reject
4	Day 0-4	Day 5-30	1.3863	0.020	Reject
5	Day 0-5	Day 6-30	1.3054	0.030	Reject
6	Day 0-6	Day 7-30	1.2550	0.035	Reject
7	Day 0-7	Day 8-30	1.2147	0.055	Fail to Reject
8	Day 0-8	Day 9-30	1.1759	0.100	Fail to Reject
9	Day 0-9	Day 10-30	1.1440	0.085	Fail to Reject
10	Day 0-10	Day 11-30	1.1198	0.100	Fail to Reject
11	Day 0-11	Day 12-30	1.1022	0.155	Fail to Reject
12	Day 0-12	Day 13-30	1.0868	0.165	Fail to Reject
13	Day 0-13	Day 14-30	1.0694	0.225	Fail to Reject
14	Day 0-14	Day 15-30	1.0580	0.210	Fail to Reject
15	Day 0-15	Day 16-30	1.0527	0.355	Fail to Reject
16	Day 0-16	Day 17-30	1.0470	0.320	Fail to Reject
17	Day 0-17	Day 18-30	1.0378	0.360	Fail to Reject
18	Day 0-18	Day 19-30	1.0425	0.375	Fail to Reject
19	Day 0-19	Day 20-30	1.0412	0.450	Fail to Reject
20	Day 0-20	Day 21-30	1.0426	0.430	Fail to Reject
21	Day 0-21	Day 22-30	1.0409	0.505	Fail to Reject
22	Day 0-22	Day 23-30	1.0253	0.505	Fail to Reject
23	Day 0-23	Day 24-30	1.0224	0.580	Fail to Reject
24	Day 0-24	Day 25-30	1.0274	0.610	Fail to Reject
25	Day 0-25	Day 26-30	1.0229	0.665	Fail to Reject
26	Day 0-26	Day 27-30	1.0204	0.685	Fail to Reject
27	Day 0-27	Day 28-30	1.0108	0.785	Fail to Reject
28	Day 0-28	Day 29-30	0.9953	0.825	Fail to Reject

Table 1: Bootstrap permutation test results for all possible split points. P-values indicate the probability of observing a variance ratio as extreme as the observed value under the null hypothesis of equal variance structure across windows.

Slice	Group 1	Group 2	Variance Ratio	P-value	Decision
29	Day 0-29	Day 30-30	1.0161	0.830	Fail to Reject

Table 1 presents the results of the bootstrap variance comparison test. For each split point, the test compares the mean column variance in the early window (Days 0 through the slice) to the mean column variance in the late window (Days slice+1 through 30). The test statistic is the ratio of early-to-late variance: values greater than 1 indicate that the early window is more variable than the late window. The p-value is computed by randomly shuffling the day columns between the two groups many times and counting how often the shuffled ratio is at least as large as the observed ratio.

Interpretation:

- **Small p-value (Reject)**: The early window has significantly higher variance than the late window. The observed concentration of variance in the early days is unlikely to have occurred by chance under the assumption of constant variance.
- **Large p-value (Fail to Reject)**: The variance in the early and late windows is similar enough that the observed ratio could easily have occurred by chance. There is no evidence of a meaningful variance difference at this boundary.
- **Transition point**: The split where the decision changes from Reject to Fail to Reject marks the boundary where the early window no longer looks meaningfully different from the late window. This transition provides a rough estimate of where elevated variance gives way to stable behavior.

It is important to note that the statistical hypothesis tested here does not perfectly match the scientific hypothesis motivating this analysis. The bootstrap test assumes a model in which variance is high for some number of days and then drops abruptly at a single changepoint. However, the variance function (Figure 2) shows that what actually happens is more gradual: variance is very large on Day 0, drops quickly over the next few days, and then settles into a stable baseline. The decline is smooth rather than a sudden jump.

This distinction matters for interpreting the results. Because the Inclan-Tiao method and the bootstrap test are both designed to detect an abrupt shift, they identify the point where the most dramatic contrast between early and late windows occurs. In practice, this captures the region where the steep decline is happening, but it does not fully describe the smooth decay pattern visible in the data. The test tells us that variance is significantly concentrated in the first few days, which is correct, but the underlying process is a rapid continuous decline rather than a single discrete break.

Despite this, the results remain informative. At early split points (slices 0-7), the test rejects the null hypothesis because the early window consists almost entirely of high-variance days near the filing date, while the late window contains the remaining stable days. The resulting variance ratio is large, and very few of our random bootstrap tests produce a ratio that extreme. As the split moves later (slices 7 and beyond), the early window begins to include enough stable-variance days that the ratio shrinks toward 1, and the test can no longer distinguish the two windows.

The transition from rejection to non-rejection at around slice 7 is not exactly consistent with the D_k changepoint estimate and the visual pattern in the variance function. This is because we need to include enough days of low variance to offset the first couple of days. This means that our test rejects up to day 7 but our strongest p-value is at the changepoint between dat 3 and 4 which agrees with our earlier changepoint tests.

Taken together, the evidence indicates that the market resolves the bulk of earnings-related uncertainty within the first three to four trading days, even though the resolution process is gradual rather than instantaneous. A more refined statistical approach—one that models a smooth variance decay rather than an abrupt shift—could provide a more precise characterization of this adjustment process, but the current results serve as a reasonable first approximation.

Conclusion

This analysis provides evidence that, even in modern financial markets, price adjustment following earnings announcements is not instantaneous. While the mean return function shows no systematic directional drift, the variance structure reveals a clear pattern of elevated uncertainty immediately following a filing, followed by a rapid transition to a more stable regime.

The combined results from the variance analysis, changepoint detection, and bootstrap testing all

point to a consistent conclusion, being that the primary adjustment period occurs within the first three to four trading days after the announcement. This is the interval during which the market processes and resolves the bulk of the information contained in the filing.

Although this adjustment window is substantially shorter than what has been documented in earlier PEAD literature, it nonetheless represents a measurable deviation from the ideal of instantaneous information incorporation. Modern markets are highly efficient, but not perfectly so.

In this sense, market efficiency is better understood as a rapid, but dynamic process, rather than an instantaneous one.

Limitations and Future Work

Several limitations should be acknowledged. First, the analysis is restricted to S&P 500 companies, which are among the most liquid and widely followed stocks in the market. As a result, the adjustment window identified here likely represents a lower bound, and less liquid firms may exhibit more pronounced effects.

Second, the analysis pools both positive and negative earnings disclosures. Separating these cases would allow for a more direct comparison with the traditional PEAD literature and may reveal additional structure in post-announcement behavior.

Third, the Inclan-Tiao method and the bootstrap test both assume an abrupt shift in variance at a single changepoint, whereas the variance function suggests a smooth, rapid decay over several days. This mismatch means the statistical hypothesis does not perfectly capture the underlying process. A more refined approach—such as fitting a parametric decay model to the variance function or using a smooth changepoint method—could better characterize the gradual transition from elevated to stable variance.

Future work could extend this analysis by examining differences across firm size, filing type, or market conditions, as well as incorporating additional data such as earnings surprises or analyst revisions to better understand the mechanisms driving the observed adjustment process.

Appendix A: Project Proposal

STAT 495R Project Proposal

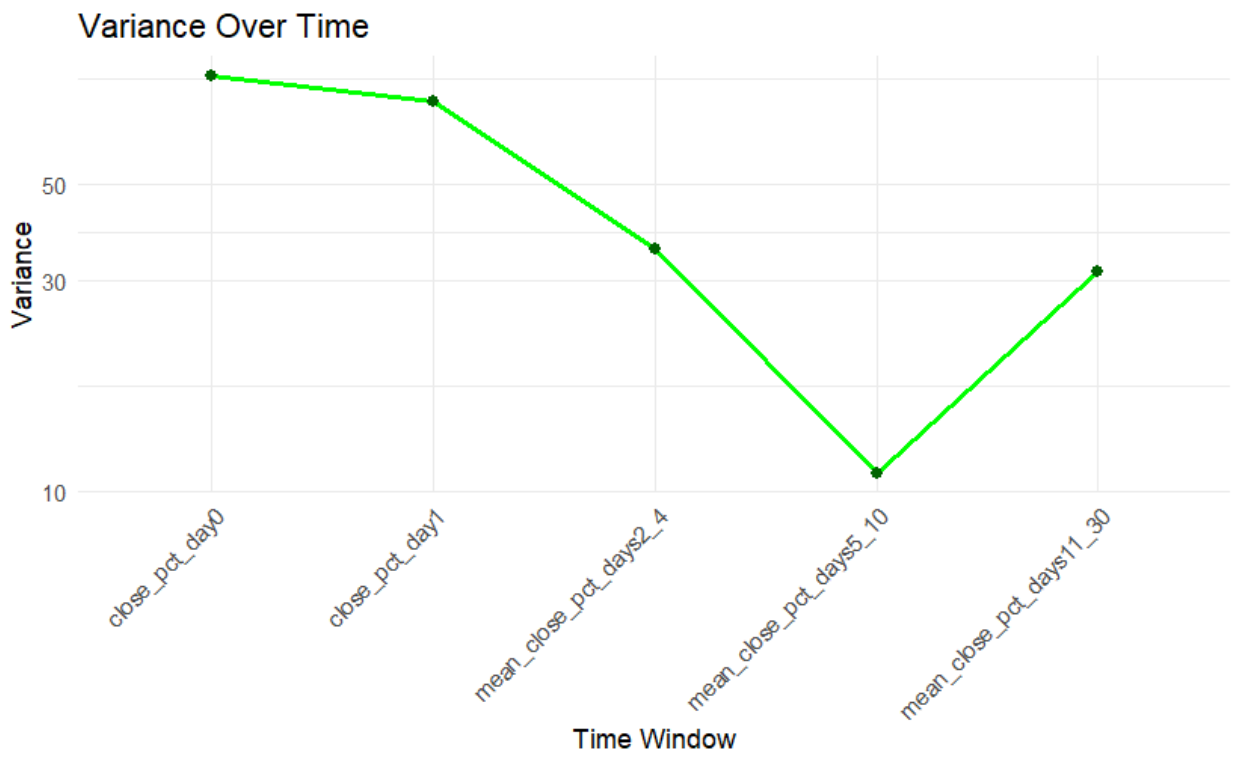
Jillian Davies & Matthew Conger

The data we are working with is derived from two main sources: first is the historic stock data from all companies on the NASDAQ that allow that data to be shared, and second, a dataset of all quarterly and annual report filing dates for all of those companies from the SEC EDGAR database. In our data, each SEC filing is the event we are looking at, and each filing date has the company indicator and 5 other variables of interest. These variables are as follows;

1. The percent change of the stock price on the filing date
2. The percent change in the stock price the day after filing
3. The mean percent change of the stock price from days 2-4
4. The mean percent change of the stock price from days 5-10
5. The mean percent change of the stock price from days 10-30

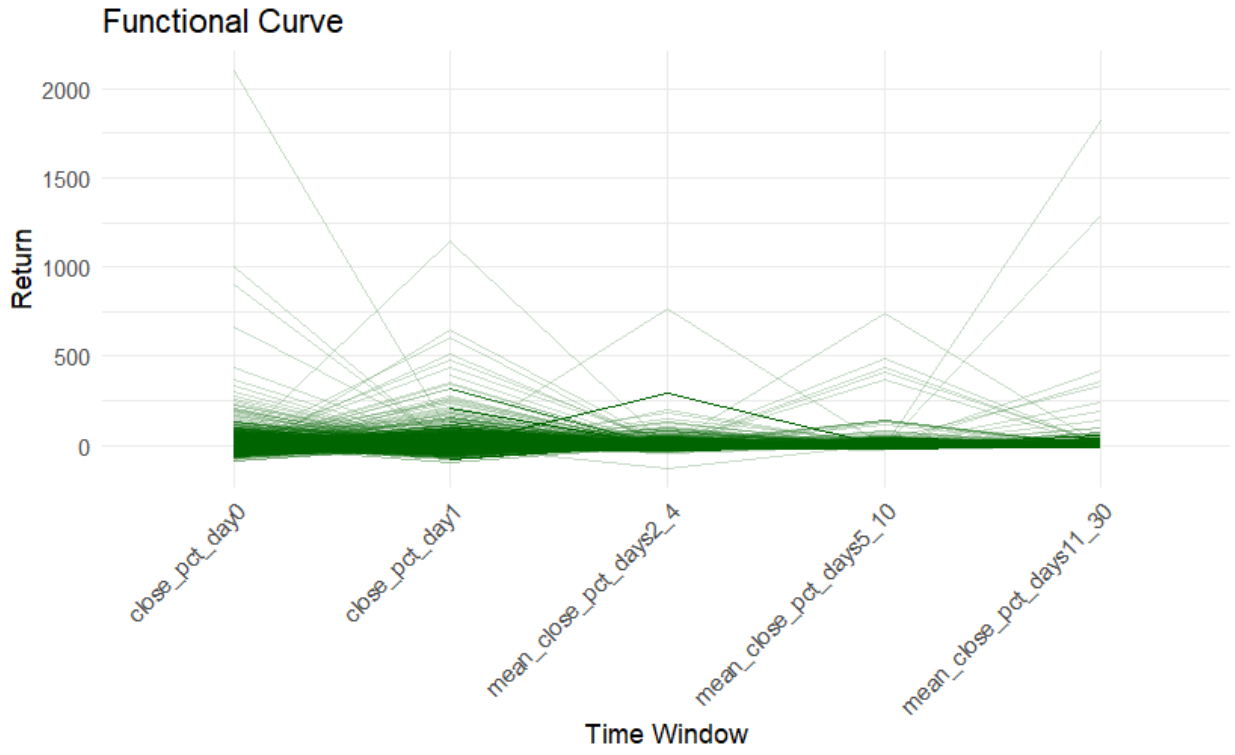
With this data, we treat each filing event as a functional observation, where the response is a curve of stock returns measured across multiple time windows after the filing event.

For more information about the data, see Data Acquisition for Reporting Analysis



This shows the variance at different points, which we would expect only to go down as we are finding the means of larger samples for some of the later points. Still, we see it jump up at the

last one. This is because the data we have tends to have a lot of large jumps and outlier values, and having a larger interval increases the likelihood of capturing extreme values within the interval.



We notice a large cluster near zero, meaning that the outliers are shrinking a large portion of our data into a tiny line, making it hard to see any potential patterns. We do notice that the dense line narrows as we increase the size of the sample, which makes sense and is what we expect to see.

These two visualizations help us understand our first steps for our analysis. These are to look into what's causing these large values and find a good way of reconciling them.

The purpose of this dataset and the overarching focus of our project is post-earnings announcement drift. The efficient market hypothesis says that when information is public, any response in the stock market to said information will occur very quickly. However, there is a lot of research showing that anomalies in stock prices can occur for days and even weeks after companies have their earnings announcements. This drift is still debated and over the past years has been thought to have shrunk by those who believe in it. We aim to obtain these insights into the subject using our data.

1. Do U.S. stock prices fully incorporate earnings-announcement information on the announcement day (day 0)?

2. Is there evidence of statistically significant post-earnings announcement drift over short windows (1–10 trading days)?
3. Has the magnitude of post-earnings announcement drift changed over time? (before 2010 vs after 2010)
4. Does the post-earnings announcement drift vary across firms or firm characteristics?
5. Does post-earnings announcement drift differ between 10-K (Annual) and 10-Q (Quarterly) filings?

Answering these questions offers two key benefits. First, it can improve decision-making for both individual and institutional investors by providing a clearer understanding of how quickly markets incorporate earnings information. Second, because firms rely on post-announcement data when making projections, any unaccounted-for post-earnings announcement drift may distort this data, potentially biasing forecasts and reducing the reliability of downstream analyses.

Appendix B: Code

```r
library(tidyverse)
library(readr)
library(dplyr)
library(lubridate)

# Load data
pead <- vroom::vroom("../Visuals and My research/daily_filing_pct_change.csv")
sp500_companies <- read_csv("../Visuals and My research/sp500_companies.csv")

# Filter to S&P 500 companies
sp500lst <- sp500_companies$Ticker
pead <- pead[pead$ticker %in% sp500lst, ]

# Date conversion and filtering
pead <- pead %>%
  mutate(filing_date = mdy(filing_date))

pead_2015_2025 <- pead %>%
  filter(between(filing_date, ymd("2015-01-01"), ymd("2025-12-31")))
# Function to prepare variance-scaled matrix for a given split
prep_mat <- function(slice = 2){
  X1 <- as.matrix(pead_2015_2025[, c(4:(slice+4))])
  X2 <- as.matrix(pead_2015_2025[, c((slice+5):34)])
  var_X1 <- apply(X1, 2, var, na.rm = TRUE)
  var_X2 <- apply(X2, 2, var, na.rm = TRUE)
  mean_var_X1 <- mean(var_X1)
  mean_var_X2 <- mean(var_X2)
  scalars_X1 <- mean_var_X1/var_X1
  Scalars_X2 <- mean_var_X2/var_X2
  X_star1 <- sweep(X1, 2, scalars_X1, `*`)
```

```r
  X_star2 <- sweep(X2, 2, Scalars_X2, `*`)
  X_star <- cbind(X_star1, X_star2)
  return(X_star)
}

# Two-sample variance comparison test for changepoint detection
bootstrap_variance_split <- function(slice, B = 200, alpha = 0.05) {
  day_cols <- paste0("day", 0:30, "_pct_change")
  X <- pead_2015_2025 %>%
    select(all_of(day_cols)) %>%
    as.matrix()
  n <- nrow(X)
  T_total <- ncol(X)
  g1_cols <- slice + 1
  X_early <- X[, 1:g1_cols, drop = FALSE]
  X_late <- X[, (g1_cols+1):T_total, drop = FALSE]
  var_early <- mean(apply(X_early, 2, var, na.rm = TRUE))
  var_late <- mean(apply(X_late, 2, var, na.rm = TRUE))
  obs_stat <- var_early / var_late
  perm_stats <- numeric(B)
  for (b in 1:B) {
    col_idx <- sample(1:T_total, size = g1_cols, replace = FALSE)
    X_perm_early <- X[, col_idx, drop = FALSE]
    X_perm_late <- X[, -col_idx, drop = FALSE]
    var_perm_early <- mean(apply(X_perm_early, 2, var, na.rm = TRUE))
    var_perm_late <- mean(apply(X_perm_late, 2, var, na.rm = TRUE))
    perm_stats[b] <- var_perm_early / var_perm_late
  }
  p_value <- mean(perm_stats >= obs_stat)
  decision <- ifelse(p_value < alpha, "Reject", "Fail to Reject")
  return(list(
```

```
    slice = slice,
    obs_stat = obs_stat,
    p_value = p_value,
    decision = decision
  ))
}

run_all_bootstrap_tests <- function(B = 200, alpha = 0.05) {
  results <- vector("list", 30)
  for (s in 0:29) {
    test_out <- bootstrap_variance_split(slice = s, B = B, alpha = alpha)
    results[[s + 1]] <- data.frame(
      slice = s,
      group1 = ifelse(s == 0, "Day 0", paste0("Day 0-", s)),
      group2 = paste0("Day ", s + 1, "-30"),
      obs_stat = test_out$obs_stat,
      p_value = test_out$p_value,
      decision = test_out$decision
    )
  }
  results_df <- do.call(rbind, results)
  rownames(results_df) <- NULL
  return(results_df)
}
set.seed(123)
bootstrap_results <- run_all_bootstrap_tests(B = 200, alpha = 0.05)
sig_rows <- bootstrap_results[bootstrap_results$decision == "Reject", ]
if (nrow(sig_rows) > 0) {
  last_sig <- sig_rows[which.max(sig_rows$slice), ]
  estimated_changepoint <- last_sig$slice + 1
} else {
  last_sig <- NULL
```

```
  estimated_changepoint <- NULL
}
day_cols <- paste0("day", 0:30, "_pct_change")
Y <- pead_2015_2025 %>%
  select(all_of(day_cols)) %>%
  as.matrix()
colnames(Y) <- 0:30

mu_hat <- colMeans(Y, na.rm = TRUE)
mu_df <- tibble(t = 0:30, mu_hat = mu_hat)

ggplot(mu_df, aes(x = t, y = mu_hat)) +
  geom_line() +
  geom_point() +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  labs(x = "Day", y = "Mean Return (%)") +
  theme_minimal()
var_hat <- apply(Y, 2, var, na.rm = TRUE)
var_df <- tibble(t = 0:30, var_hat = var_hat)

ggplot(var_df, aes(x = t, y = var_hat)) +
  geom_line() +
  geom_point() +
  labs(x = "Day", y = "Variance") +
  theme_minimal()
a2 <- var_hat
Ck <- cumsum(a2)
T_total <- length(a2)
Dk <- Ck / Ck[T_total] - (1:T_total)/T_total
k_hat <- which.max(abs(Dk))
day_hat <- (0:30)[k_hat]
```

```r
dk_df <- tibble(k = 1:T_total, day = 0:30, Dk = Dk)


ggplot(dk_df, aes(x = day, y = Dk)) +
  geom_line() +
  geom_point() +
  geom_point(data = dk_df[k_hat, ], aes(x = day, y = Dk),
             shape = 18, size = 4, color = "red") +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(x = "Day", y = expression(D[k])) +
  theme_minimal()
library(knitr)


bootstrap_results %>%
  mutate(
    obs_stat = round(obs_stat, 4),
    p_value = round(p_value, 3)
  ) %>%
  select(Slice = slice, `Group 1` = group1, `Group 2` = group2,
         `Variance Ratio` = obs_stat, `P-value` = p_value, Decision = decision) %>%
  kable(booktabs = TRUE, longtable = TRUE)
```